

Detection Testing & Validation Report

Synthetic test cases used to validate SPL logic and expected analyst outcomes

Test ID	Detection	Input condition	Expected result	Observed portfolio result
T-01	Brute Force	14 failures from one source/user in <5 min	Alert fires Medium/High depending threshold	Triggered
T-02	Brute Force	3 failures over 20 min	No alert	Not triggered
T-03	Password Spray	1 failure each across 5 users from one source in <10 min	High alert	Triggered
T-04	Credential Stuffing	12 failures across 4 users + one success	High alert	Triggered
T-05	Failed -> Success	Failure sequence and success same user/source <10 min	High alert	Triggered
T-06	Impossible Travel	Same managed device through known VPN egress	Should be reviewed as likely benign after enrichment	Detection fires; analyst disposition benign
T-07	Impossible Travel	Different geography + unknown device + password-only within 32 min	High alert	Triggered / escalated
T-08	Suspicious PowerShell	Approved signed deployment script	May match keyword logic; analyst/tuning should suppress approved context	Benign positive
T-09	Suspicious PowerShell	Word -> encoded PowerShell + outbound connection	High alert	Triggered / escalated

Validation Findings

- Authentication detections successfully separate high-volume single-account guessing from multi-account spraying patterns.
- Sequence-based correlation improves prioritization because a successful login after repeated failures is higher risk than failures alone.
- Impossible-travel logic requires enrichment; VPN/device context is necessary to prevent geography-only over-escalation.
- PowerShell keyword logic is intentionally broad and must be enriched with parent process, signature, script path, user context, and network behavior.
- Production implementation should use environment-specific baselines, data-quality monitoring, allowlists with owners/expiration, and change control.